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Hull hydrodynamic optimization of autonomous underwater vehicles operating at snorkeling depth

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ABSTRACT

Traditionally autonomous underwater vehicles (AUVs) have been built with a torpedo-like shape. This common shaping is hydrodynamically suboptimal for those AUVs required to operate at snorkeling condition near the free surface. In this case, the wave resistance associated to the wavy deformation of the sea surface induced by the motion of the platform is an important component of the drag. This work has investigated the optimum hull shape of an underwater vehicle moving near the free surface. Specifically a first-order Rankine panel method has been implemented to compute the wave resistance on a body of revolution moving close to the free surface. A simulated annealing algorithm was then employed to search those set of parameters defining the hull shape that minimize the wave resistance. The optimization was constrained to keep constant the total volume of the vehicle. The total drag of scaled models of the torpedo-like and resulting optimum shapes was measured in the naval tank of the University of Trieste. Measurements showed a smaller resistance of the optimized shape in the range of the considered Froude numbers.

1. Introduction

Autonomous underwater vehicles (AUVs) are submarine robots able to command position and control mission plan to carry out underwater samplings autonomously (Curtin et al., 1993). Their hydrodynamic shape, electrical propulsion, submarine navigation and positioning allow continuous sampling of underwater conditions. They can integrate different measurement instrumentation (Bales and Levine, 1994) and their speed range from 1 to 5 knots. AUVs constitute a notorious advance in observing and experimenting in the marine environment (An et al., 2001). A wide variety of underwater data collection missions are well suited for these underwater robots. For example, AUVs have been employed to sample coastal fronts (Nadis, 1997), to monitor coastal areas (An et al., 2001), to measure turbulence in the thermocline (Dhanak and Holappa, 1996), to obtain interdisciplinary data (Millard et al., 1998), to do fisheries research (Gaudey and Bellingham, 1994) or to install submarine cables under frozen seas (MacFarlane, 1997).

Transformation of individual AUV technologies into distributed ocean observing systems is underway. The idea is to collect data with a network of interactive and cooperative AUVs. These adaptive and networked systems can perform synoptic surveys of large volumes of water minimizing aliasing of temporal and spatial variability. It has been suggested that to realize an affordable and deployable network, a new class of AUVs must be designed with concepts and constraints not necessarily applied in developing solo AUVs (Curtin et al., 2005). These network class AUVs must show a higher degree of specialization in the design to optimize the cost-effectiveness of the overall network. Curtin et al. (2005) have proposed a broad segmentation of network class AUVs based on their operating regime: bottom layer, interior and surface layer AUVs. Bottom layer AUVs would serve in the network as a transport, tanker and relative navigation reference. They must operate as AUV carriers for long transits and as mostly stationary, bottom dwelling energy providers and location references. Interior class AUVs would constitute the expeditionary platforms of the network. These vehicles must integrate different sensors and be capable of underwater navigation and communication. Finally, surface layer AUVs would constitute the control nodes. They would act as communication links between the acoustic transmissions of the interior and bottom layer AUVs and the above surface radio transmissions. They would also provide geo-coordinates and time references from the global positioning system (GPS) to the interior and bottom layer AUVs.

The design of interior network class AUVs is the most mature. This is because they are similar to existing AUVs and thus they

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share the same design criterion (Griffiths, 2003; Bertram and Alvarez, 2006). The least developed design concerns bottom layer network class AUVs. Concerning surface layer AUVs, several prototypes exist today. The SAUV II (Jalbert et al., 2003), Folaga (Alvarez et al., 2005) and Cormorán (Martínez-Ledesma et al., 2004) are primarily surface layer AUVs. The SAUV II is a solar powered AUV designed for long endurance missions where real time bi-directional communications to shore are critical. Folaga and Cormorán are vehicles specifically designed for oceanographic sampling. Both they move submerged near the free surface until a measurement point is reached. Then, they perform a vertical dive to sample the water column. These vehicles operate at snorkelling condition to keep the snorkel (with the radio and GPS antennae) above the sea surface. This specific operational procedure introduces important constraints in the hydrodynamic design of these vehicles. Unlike the interior and bottom layer AUVs, a substantial part of the resistance experienced by these vehicles is not of frictional nature but due to the dissipation of motion energy to distort the free surface. The effect of this dissipation is the generation of a characteristic wave pattern behind the vehicle. This component of the total resistance is named wave resistance. Thus, an appropriate hydrodynamic design of these network class vehicles to reduce the wave resistance would be desired.

Hydrodynamic optimization of hull-forms is rarely used in naval architecture because existing design tools are not sufficiently robust and/or fast to be used within an optimization scheme. For this reason, ship hull-form design typically amounts to selection among a handful of alternatives that are deemed reasonable based on previous experience. However, some works have considered the problem of determining optimized hull-forms taking into account the total resistance (friction+wave). Hendrix et al. (2001) has proposed a simple CFD tool to optimize monohull ships for which the total resistance is minimized. The friction resistance is estimated using the classical International Towing Tank Conference (ITTC) friction-drag formula (van Manen and van Oossanen, 1988) while the wave resistance is obtained using the zeroth-order slender-ship approximation (Noblesse, 1983). More recently, Subramanian and Joy (2004) have considered the problem of optimizing the total resistance of catamarans. Again, the ITCC friction-drag formula is employed to estimate the friction resistance while the wave resistance component is obtained using Michell's theory for slender vessels with a modification using a wave interference factor (Tuck, 1987). Unfortunately, classical methods (thin ship theories, slender-body theories) to compute wave resistance introduce simplifications which imply limitations regarding the AUV's geometry. Real AUV geometries are generally not thin or slender enough. The differences between computational and experimental results could be consequently unacceptable.

Practical predictions of wave resistance for general hull geometries are based almost exclusively on boundary element methods. These boundary element methods represent the flow as a superposition of Rankine sources and sometimes also dipoles or vortices. Concerning the free-surface boundary condition, linearizing the differences between the actual flow and uniform flow to simplify the nonlinear boundary condition is usually employed. This simplification is known like Kelvin condition. This crude approximation is for practical purposes no longer accepted. Dawson (1977) proposed to use the potential of a double-body flow and the undisturbed water surface as a better approximation. The double-body linearization was popular until the early 1990s (e.g. Aanesland, 1989; Nakos, 1990; Raven, 1990; Lalli et al., 1992). Nowadays, boundary element methods considering fully nonlinear boundary conditions are the state of art to compute wave resistance (Ni, 1987; Jensen, 1988; Kim and Lucas, 1990, 1992; Raven, 1992, 1996; Janson, 1996).

This work considers the hydrodynamic optimization of hull geometries of surface layer AUVs. The main objective is to find an appropriate hull shape to reduce the total resistance (friction+wave) of a vehicle moving submerged near the free surface. The optimization process is constrained to keep the volume of the vehicle constant. Unlike previous works, a boundary element method considering fully nonlinear boundary conditions is applied to compute the wave resistance component. The article is divided in five sections. The mathematical formulation of the wave resistance problem is briefly reviewed in Section 2. Section 3 details the numerical implementation of the optimization problem. Numerical results are shown and compared with experimental measurements in Section 4. Finally, Section 5 discusses and concludes the work.

2. Mathematical formulation of the hull resistance

Traditionally, the total hull resistance is split up into friction, form and wave resistance components. The friction resistance is due to viscosity. Water particles have the same speed that the body near its hull while at a short distance away from it, fluid particles move with the speed of the outer flow. The transition region between the hull and the outer flow forms the boundary layer. Velocity changes across the boundary layer induce shear stresses that integrated over the wetted surface yield the friction resistance. Accurate computation of friction resistance would require an enormous computational effort incompatible with an optimization process. For this reason, friction resistance is usually approximated by the drag generated by a turbulent layer flow over a flat plate with the same wetted area and length as the body. The formulation employed in this work is the one accepted as a standard by the ITTC (van Manen and van Oossanen, 1988):

$$C_F = \frac{0.075}{(\log_{10}(Re) - 2)^2} \tag{1}$$

where C_F is the nondimensional friction-drag coefficient and Re is the Reynolds number based on the body length.

The form resistance is associated to the geometry of the body. The particular form of the hull induces average shear stresses, energy losses in the boundary layer, vortices and flow separation preventing an increase to stagnation pressure in the aftbody as predicted in an ideal fluid theory. A pressure gradient opposing to the motion results between the fore and aftbody. It is generally assumed that the form resistance is proportional to the friction resistance with a proportionality factor k (known like form factor) given by (Hendrix et al., 2001)

$$k = 0.6\sqrt{\frac{D}{L^3}} + 9\frac{D}{L^3} \tag{2}$$

where L and D stand for the length and displacement volume of the body, respectively. Previous works have applied Eqs. (1) and (2) to estimate the friction and form resistances of AUVs (Barros et al., 2006; Phillips et al., 2007). This is justified by the fact that laminar flows are not very likely to occur over the AUV surface in real sea water due to environmental turbulence. Experimental results reported good agreement between the ITTC-57 model and measurements of friction resistances for different kind of surfaces and for Reynolds numbers ranging from 10^6 to 5×10^6 (Candries et al., 2001). This is the interval of Reynolds numbers where most AUVs operate, including the one considered in this study ($Re\sim1.5\times10^6$). Form factors depending on the thickness ratio have been proposed by Horner (1965) for streamlined bodies.

Finally, the wave resistance is estimated considering the steady motion of a body in initially smooth water assuming an ideal fluid, i.e. especially neglecting all viscous effects. The body moves with constant speed $\vec{V}$ in water of infinite depth, submerged at constant depth near the free surface of the water. The following physical simplifications are assumed: water is incompressible, irrotational, and inviscid; surface tension is negligible; there are no breaking waves; the hull has no knuckles which cross streamlines and appendages and propellers are not included in the model. Under these assumptions the velocity field of the fluid can be described, in a inertial coordinate frame attached to the vehicle, in terms of a potential function ϕ with $\vec{v}(x,y)=\nabla\phi$. For the considered ideal flow, continuity gives Laplace's equation for the potential which holds in the whole fluid domain:

$$\nabla^2\phi = 0 \tag{3}$$

All equations are formulated here in a right-handed Cartesian coordinate system with x pointing forward towards the bow and z pointing upward. The moment about the y-axis is positive clockwise.

A unique description of the problem requires further conditions on all boundaries of the fluid. At the hull water does not penetrate the body's surface. The hull condition requires that the normal velocity on the hull vanishes:

$$\vec{n}\nabla\phi = 0 \tag{4}$$

$\vec{n}$ is the inward unit normal vector on the ship hull. The kinematic free-surface condition at the water surface $z=\zeta$ establishes that a water particle at the free surface must always remain on it:

$$\nabla\phi\nabla\zeta = \phi_z \tag{5}$$

where for simplification, we write $\zeta(x,y,z)$ with $\zeta_z=0$. A dynamic boundary condition also holds at the free surface:

$$\frac{1}{2}(\nabla\phi)^2 + gz = \frac{1}{2}|\vec{V}|^2 \tag{6}$$

with $g=9.8\,\mathrm{ms}^{-2}$. Remaining boundary conditions applied to the wave pattern generated at the free surface by the motion of the body. Specifically, a radiation condition imposes that waves generated by the body do not propagate ahead. A decay condition implies that the flow is undisturbed far away from the body. Finally, open-boundary conditions are considered at any artificial boundary of the computation domain.

Once a potential has been determined, the resistance force can be determined by direct pressure integration on the hull:

$$f_1 = \int_S P n_x \mathrm{d}S \tag{7}$$

S is the wetted surface. P is the pressure determined from Bernoulli's equation.

$$P = \frac{1}{2}\rho(|\vec{V}|^2 - (\nabla\phi)^2) \tag{8}$$

ρ is the density of water. The force in x-direction, f_1, is the negative wave resistance. The nondimensional wave resistance coefficient is

$$C_W = \frac{-f_1}{(1/2)\rho|\vec{V}|^2 S} \tag{9}$$

Once computed all the resistance components, the nondimensional total resistance coefficient is given by

$$C_T = C_W + (1+k)C_F \tag{10}$$

3. Numerical implementation of the optimization problem

Axisymmetric bodies moving submerged near to the free surface are considered in this work. The geometry of revolution is parametrized by three lengths L_a, L_f, R, for aft, front part and radius at the bodies centre and two exponents n_a and n_f (Fig. 1). The aft part and the front part radii follow from

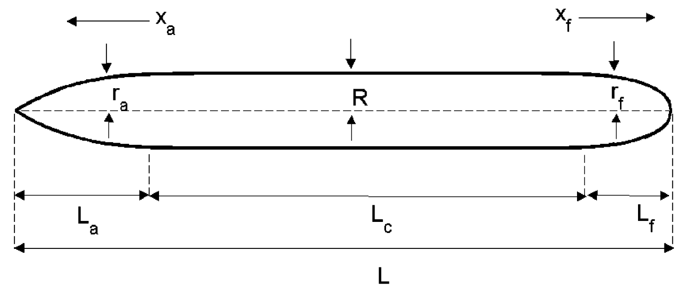

Fig. 1. Parametrization of the hull geometry.

$$r_a = R\left(1-\left(\frac{(L_a-x)}{L_a}\right)^{n_a}\right) \quad r_f = R\left(1-\left(\frac{(x-L_a-L_c)}{L_f}\right)^{n_f}\right)^{1/n_f} \tag{11}$$

$r_{a(f)}$ is the radius at the aft (front) position x and R is the radius of the central cylinder. The central part of the body is considered cylindrical with a length, L_c, obtained from a volume constraint. We neglect for the time being the existence of a snorkel, assuming no interaction between the wave systems of main body and the snorkel. This assumption may be justified, as the Froude numbers for main body and snorkel are different by two orders of magnitude.

An iterative procedure is employed to solve the nonlinear wave resistance problem (Bertram, 2000). This resolution procedure seems the most adequate considering that a nonlinear boundary condition exists on the free surface and the free-surface position is not a priori known. Specifically, the potential ϕ is approximated by an arbitrary function Φ and the free-surface ζ by an arbitrary approximation Z. Combining the dynamic and kinematic boundary conditions and linearizing consistently around the approximations, the nonlinear free-surface condition yields at $z=Z$ (Jensen, 1988; Bertram, 2000):

$$\begin{aligned}&2(\mathbf{a}\cdot\nabla\phi + \Phi_x\Phi_y\phi_{xy} + \Phi_x\Phi_z\phi_{xz} + \Phi_y\Phi_z\phi_{yz})\\&+\Phi_x^2\phi_{xx} + \Phi_y^2\phi_{yy} + \Phi_z^2\phi_{zz} + g\phi_z - B\nabla\Phi\nabla\phi =\\&2\mathbf{a}\cdot\nabla\phi - B(|1/2((\nabla\phi)^2+v^2) - gZ)\end{aligned} \tag{12}$$

with $\mathbf{a}=1/2\nabla((\nabla\Phi)^2)$ and $B=(\mathbf{a}\cdot\nabla\Phi+g\Phi_z)_z/(g+a_3)$, being a_3 the third component of vector $\mathbf{a}$. This condition is rather complicated involving up to second derivatives of the potential, but it can be simply repeated in an iterative process which is started with uniform flow ($\Phi=\{-Vx,0,0\}$) and no waves ($Z=0$). In each iterative step, wave elevation, potential, and position are updated yielding successively better approximations for the solution of the nonlinear problem, if the process converges. Convergence is usually rapid. Typically 3 or 4 iterations suffice.

The free surface and the body are discretized using classical first-order Rankine panels as proposed by Hess and Smith (1962, 1964). Quadrilateral panels are used to discretize the free surface and central part of the body while triangular panels are employed to represent aft and frontal corners. Desingularized Rankine point sources are used above the free surface to compute second derivatives of the velocity potential. The desingularization distance is twice the grid spacing in x-direction. This distance was found to give more reasonable results than the dimensionally inconsistent recommendation of Beck et al. (1999). During the iteration, the collocation points at the free surface are updated, but the position of the sources remains unchanged. Mirror images of panels are used in y direction with respect to $y=0$. The decay condition—like the Laplace equation—is automatically fulfilled by all elements. The radiation condition and the open-boundary condition are fulfilled by adding an extra row of source elements at the downstream end of the computational domain and an extra row of collocation points at the upstream end (Jensen, 1988); Thiart and Bertram, 1998). For equidistant grids this can also be interpreted as shifting or staggering the grid of collocation points versus the grid of source elements. This technique shows absolutely no numerical damping or distortion of the wave length, but requires all derivatives in the formulation to be evaluated numerically.

A simulated annealing (SA) technique (Kirkpatrick et al., 1983) was employed to solve the optimization problem. SA exploits the analogy between the way a metal cools and freezes into a minimum energy crystalline structure (annealing process) and the search for a minimum in more general problems. The procedure employs a Metropolis algorithm (Metropolis et al., 1953) to explore the space of solutions. The random algorithm not only accepts trades that decrease the objective function, but also some changes that increase it. The latter are accepted with a probability $e^{(-\Delta/T)}$, where Δ is the increase in the value of the cost function and T is a control parameter called temperature of the system. This parameter is sequentially reduced from an initial value, after making many trades and observing a slow decline in the cost function. The algorithm stops when a specified final value of the temperature is reached. SA's advantage over other methods is an ability to avoid becoming trapped at local minima. The SA optimization process is summarized in the following pseudocode:

- Set parameters $Q = [L_a, L_f, R, n_{a,}, n_f]$ to the values encoding the geometry of the original hull
- Set temperature T to T_{init}, scale parameter ε (<1) and minimum temperature T_{min}
- Repeat
  - i. Perturb $Q = [L_a, L_f, R, n_{a,}, n_f]$ randomly to obtain a new set of parameters $Q_{new} = [L_a, L_f, R, n_{a,}, n_f]_{new}$
  - ii. Compute $\Delta = \psi(Q_{new}) - \psi(Q)$ being the cost function the computed total resistance $\psi = 1/2 C_T |\vec{V}|^2 S$
  - iii. Compute the probability $P = \min(1, e^{(-\Delta/T)})$
  - iv. If(random()$<P$) $Q = Q_{new}$
  - v. Decrease T, $T = \varepsilon T$
- Until ($T < T_{min}$)
- Return Q

The overall numerical procedure was implemented in Matlab. The advantage of Matlab is an easy visualization without need of external software. The computational time to compute the wave resistance of a hull shape is typically 40 s on a Pentium IV processor machine of 3.06 GHz for a grid of 1000 elements (unknowns).

4. Numerical results

4.1. Model validation

The computation of the wave resistance experienced by an elongated spheroid moving submerged near the free surface has been done to validate the developed model. This is a standard case employed to compare the performance of different wave resistance models. The spheroid has an aspect ratio1:5 and a draught $T = 0.245$ L, where L is the length of the spheroid. Our results have been compared with previous computations based on two different Rankine panel methods, Bertram et al. (1991). The numerical model has been implemented discretizing the body with 1127 panels on one half of the hull and a cosine law partition. The free surface was discretized with 1037 panels. Fig. 2 compares the nonlinear results for wave drag with results published by Bertram et al. (1991). The agreement is very good showing that the method was correctly implemented.

4.2. Hydrodynamic optimization

As a particular example, the hull optimization of the vehicle Cormoran is now considered. Cormoran is a surface layer AUV with a torpedo-like geometry defined by $L_a = 0.38$ m, $L_f = 0.24$ m, $R = 0.08$ m, $n_a = 3$ and $n_f = 2.3$. The total displacement volume and wet surface of the vehicle are 0.0245 m^3 and 0.63 m^2, respectively. The nominal speed of this vehicle is around 1 m/s and its draught 0.1 m. The hull has been optimized for minimum total resistance averaged for 0.9, 1, and 1.1 m/s design speed, corresponding to Froude numbers ranging from 0.26 to 0.3. The total resistance is computed as sum of friction, form and wave resistance. Friction and form resistances were estimated using Eqs. (1) and (2) while wave resistance was computed with the Matlab wave resistance code described above. A total of 863 source elements were distributed on the body hull (3 4 3) (Fig. 3a) and free surface (5 2 0), employing symmetry in y by mirror images of the elements. This mesh represents a good compromise between accuracy and computing effort. Providing six to eight grid points per wavelength, it guarantees a proper representation of the wave pattern (Peterson and Middleton, 1963; Bertram, 2000). Constraints for the optimization were constant displacement volume and maximum length of the vehicle of 1.5 m. The latter prevents the generation in the optimization process of long and thin geometries, unfeasible to implement in real AUVs due to payload size and portability restrictions.

Optimization parameters were $T_{init} = 120$, $T_{min} = 10^{-5}$ and $\varepsilon = 0.98$. The optimization required a total of 808 evaluations of the cost function leading to a total computation time of 15 h. Table 1 and Fig. 3b summarize the results of the optimization.

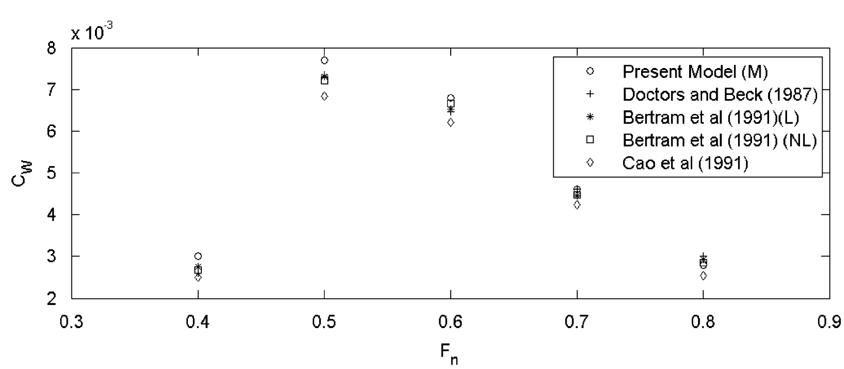

Fig. 2. Comparison of the results of different numerical models found in the literature with the developed method to compute the wave resistance of a submerged spheroid moving near the free surface (Cao et al., 1991; Doctors and Beck, 1987).

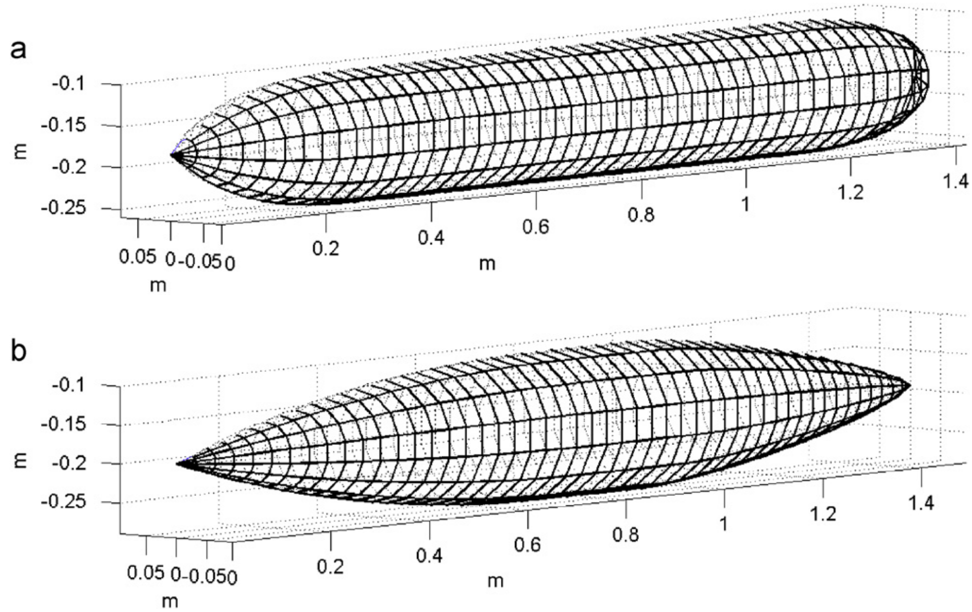

Fig. 3. Hull geometry of (a) Cormoran and (b) the resulting optimized hull.

Table 1

	Original	Optimized
n_a	3	2
n_f	2.3	1.3
Total length (m)	1.42	1.49
Length of aft part (m)	0.38	0.73
Length of forward part (m)	0.24	0.54
Radius (m)	0.08	0.095
Wet surface (m^2)	0.63	0.62
Volume (m^3)	0.0245	0.0245
Viscous resistance (N)	1.614	1.541
Wave resistance (N)	0.418	0.006
Total resistance (N)	2.03	1.547

Fig. 4a and b compare the surface wave patterns generated by the original and optimized hulls. A substantial reduction of the wave height is observed in the case of the optimized hull, indicating a lower wave resistance. Specifically, the maximum wave amplitude in the wave patterns of the original and optimized shapes are 0.02 and 0.012 m, respectively. This is supported by Fig. 5(a) and (b) where the total, friction, form and wave resistances are plotted for different Froude numbers for the original and optimized geometries, respectively. Friction resistance dominates in both cases. In the original hull shape, wave resistance is superior to form resistance except in a small interval around the Froude number 0.285. It ranges from 2% up to 30% of the total resistance in the considered interval of Froude numbers. Conversely, wave resistance is negligible for the optimized shape.

5. Experimental validation

Total resistances of the original and optimized hulls have been measured in the ship model basin of the University of Trieste in Italy. Two models of the original and optimized hulls were built using fiber-glass and foam with a scale 2:1 (Fig. 6a and b). Models were rigidly connected to the carriage through a connection plate. This kind of connection is usual in towing tank experiments. Vertical force due to the model buoyancy is ignored by the carriage load cell, measuring only the horizontal component of the towing force. Turbulence stimulators made by wire of 0.0008 m were added at the forebody of each model. Total resistances were measured at velocities ranging from 0.5 up to 1.05 m/s with speed increments of 0.05 m/s. The draught was 0.05 m correspondingly to the model scale. Resistance of the connection plate was also measured for the same range of speeds, in order to deduct its effect on the resistance of the models. Connection plate resistance was of the same order of magnitude than the accuracy of the measuring system (0.04 N) and so it was neglected during the data processing.

Fig. 7 compares the nondimensional total resistance coefficient obtained from the experimental measurements and the numerical model for the original and optimized shapes. Results indicate that the numerical model underestimates the total resistance in both cases. Surprisingly, the differences found between measured and computed values are not too big considering the rough models employed to estimate friction and form resistance coefficients. While discrepancies exist in the absolute magnitude of the total resistance, its behaviour versus the Froude number is well captured by the numerical approach for the case of the optimized hull. Bigger disagreements are found between the measurements and the numerical results concerning the behaviour of the total resistance in the case of the original hull. This disagreement results into a more restrictive solution of the numerical code, in the sense that measurements show that the optimized hull experiments a lower drag than the original hull up to Froude numbers of 0.33. Measurements and numerical results agree in the fact that the optimized hull shows lower resistance than the original shape in the range of Froude numbers where the numerical optimization was performed.

6. Discussion and conclusion

This work has considered the optimization of the hydrodynamics of the hull of surface layer AUVs. The role of these AUVs in an observing network is to provide communication link and absolute positioning to interior and bottom layer AUVs. Thus, they must operate near the free surface to keep the communication and positioning snorkel above the sea level. This operational

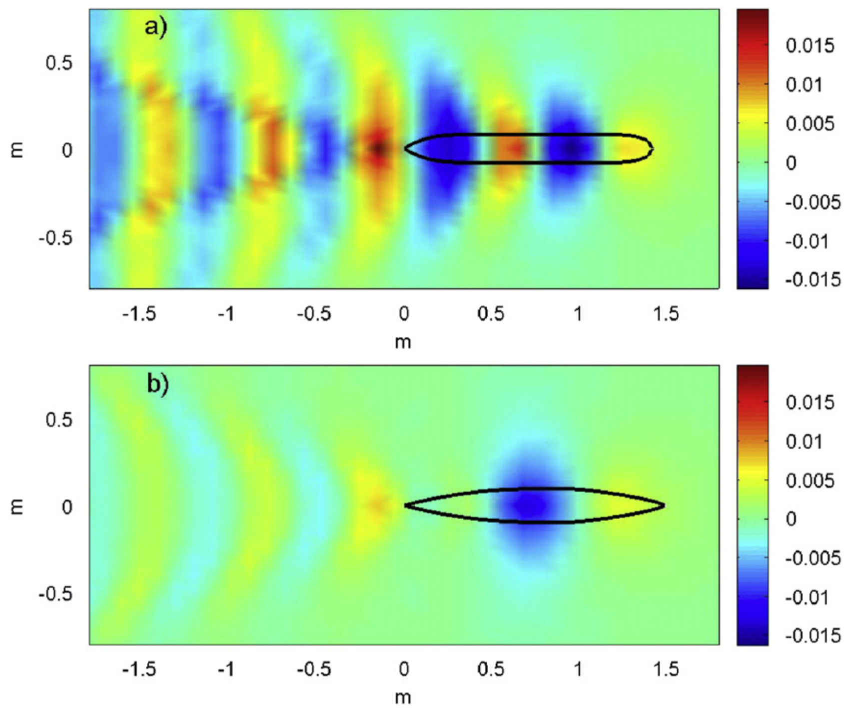

Fig. 4. Surface wave pattern generated by (a) the original and (b) optimized hulls at $F_n = 0.26$.

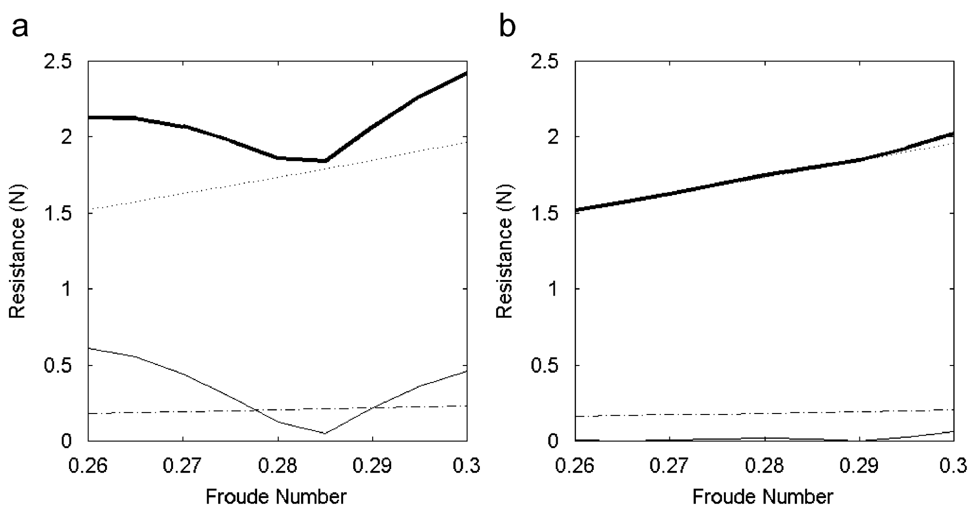

Fig. 5. Total (solid thick line), friction (dotted line), form (dashed-dotted line) and wave (thin solid line) resistances for the (a) original and (b) optimized hull geometries.

specialization translates into design requirements usually not applied to other more general AUVs.

Optimizing the hull geometry to reduce the total resistance is a fundamental aspect to increase the vehicle autonomy at sea. The resistance to the motion experienced by an AUV operating near the free surface is bigger than operating under the same conditions in the interior ocean. The extra resistance component, named wave resistance, is generated by the dissipation of kinetic energy in form of radiating free-surface waves. Wave resistance can become a substantial part of the total resistance. A three dimensional Rankine panel method has been programmed in this work to compute the wave resistance experienced by a surface layer AUV. The approach takes into account the nonlinear character of the boundary condition at the free surface. Friction and form resistances were estimated using the ITTC-57 model. This is a rough model where the resistances depend on the displacement, length and Reynolds number of the vehicle. More sophisticated approaches could be considered to represent the viscous drag. Unfortunately, this would increase the computational effort making the optimization unfeasible. To explore the benefits of the hydrodynamic hull optimization in surface layer AUVs, the optimization of an existing AUV designed to operate near the free surface was considered. The AUV called Cormoran has a typical torpedo shape, commonly employed in AUV building. The hull shape resulting from the optimization process substantially reduces (up to 25%) the estimated total resistance. This reduction mostly applies to the wave resistance component as it is shown looking at the amplitude of the radiated wave pattern. This could be expected a priori. Both hulls, original and optimized, have the same displacement and similar Reynolds number and length. Thus, the ITTC-57 model provides for both similar values for the friction and form resistances.

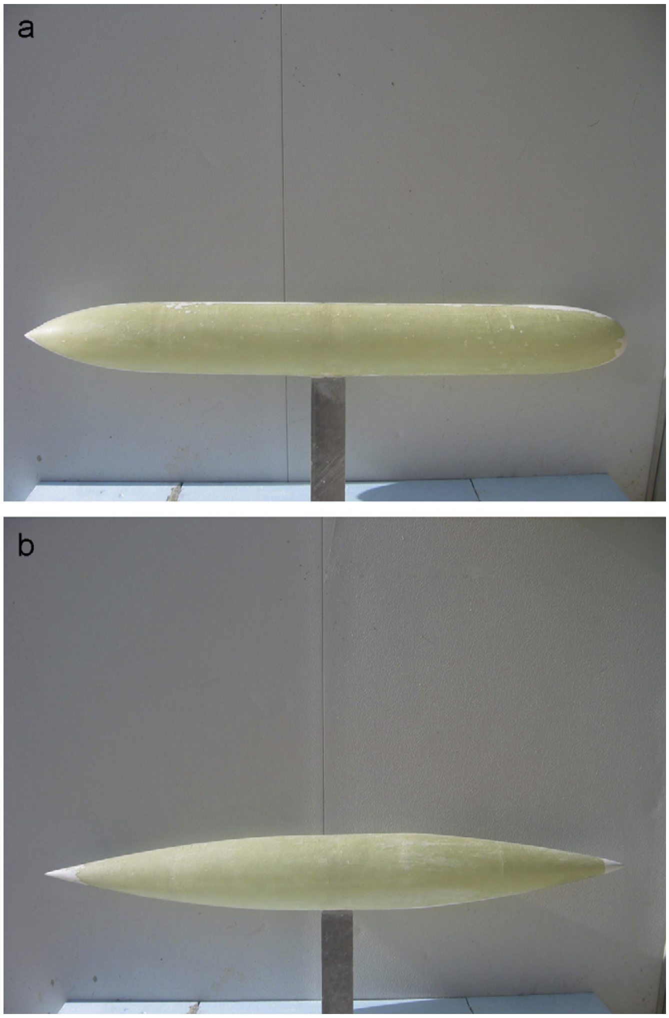

Fig. 6. Models of the original (a) and optimized (b) hulls.

Experimental verification of the numerical results is always required when designing new hull geometries. In the present case, resistance measurements were done with models of the original and optimized hulls. Experimental results show in general that the resistance values estimated by the proposed model are not too far from reality. In the case of the optimized hull, numerical results nicely reproduce the observed resistance curve. This is characterized by a plateau region of the resistance for Froude numbers smaller than 0.3 followed by an exponential growth of the resistance as the Froude number is increased. Conversely, a clear disagreement has been found between the observed and estimated resistance curves in the original torpedo-like configuration. The origin of the different performance of the numerical model to estimate the total resistance in the torpedo-like and optimized shapes is hypothesized to arise in the viscous resistance component. The optimized hull is streamlined without abrupt changes in curvature, so flow separation is unexpected. The friction resistance is well approximated by the ITCC-57 model. A more complex situation is expected for the torpedo-like geometry. The body is not streamlined and flow separation is likely to occur. Viscous drag is more relevant than in the previous case but its complexity is not well represented by the ITCC-57 model.

Concluding, it should be pointed out that expansion of the technological applications of AUVs requires a higher degree of specialization of these autonomous vehicles. This specialization requires adequate hydrodynamic designs in order to increase the autonomy and/or performance of the vehicle. In the case of surface layer AUVs, an optimized design of the hull can represent a substantial increase of the autonomy of the vehicle. This optimization process can be carried out using numerical models to estimate friction, form and wave resistances. The wave resistance component is well represented by a three dimensional Rankine panel method that considers the nonlinear conditions at the free surface. However, comparison with experiments indicates that more sophisticated models should be considered to treat the resistance components with viscous origin.

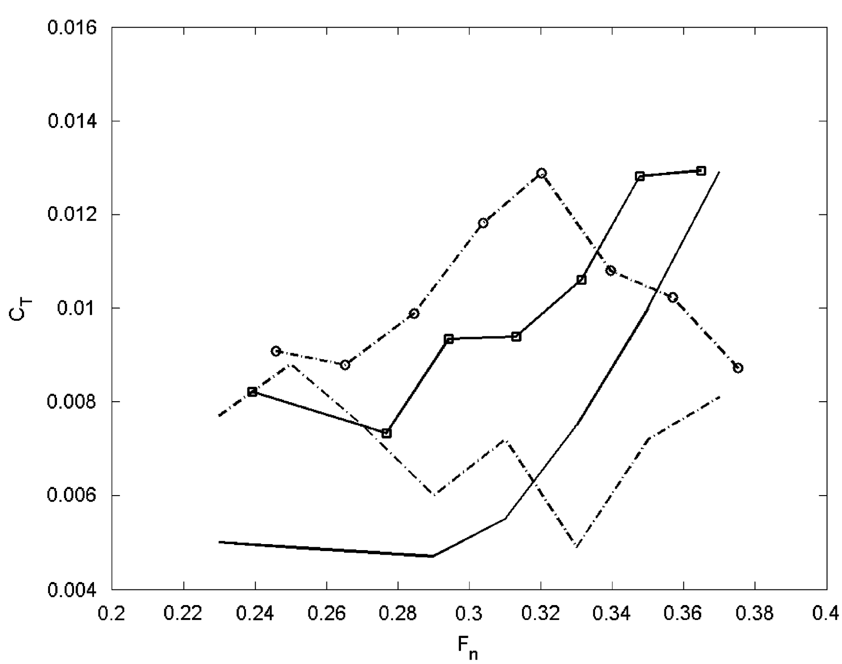

Fig. 7. Nondimensional total resistance coefficient obtained from the experimental measurements (dashed line with circles and solid line with squares for the original and optimized hulls, respectively) and the numerical model (dashed line and solid line for the original and optimized hulls, respectively).

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